

**MPHARM**  
**(SEM I) THEORY EXAMINATION 2025-26**  
**PHYTOCHEMISTRY**

**TIME: 3 HRS**

**M.MARKS: 75**

**Note:** Attempt all Sections. If require any missing data; then choose suitably.

**SECTION A**

**1. Attempt all questions in brief.**

**2 x 10 = 20**

|    |                                                                       |
|----|-----------------------------------------------------------------------|
| a. | Draw the chemical structure of citral & menthol.                      |
| b. | Define quinine & vinca.                                               |
| c. | Write the principle of microwave extraction.                          |
| d. | Define Lead compounds.                                                |
| e. | Give the importance of caffeine & carvone.                            |
| f. | Draw a chemical structure of luteolin.                                |
| g. | Mention the characterization & purification technique for terpenoids. |
| h. | Give the structure & chemical constituent of piperine & taxol.        |
| i. | Write the principle of HPLC.                                          |
| j. | Write the principle of extraction.                                    |

**SECTION B**

**2. Attempt any two parts of the following:**

**10 x 2 = 20**

|    |                                                                                                                                               |
|----|-----------------------------------------------------------------------------------------------------------------------------------------------|
| a. | Discuss the flash column chromatography.                                                                                                      |
| b. | Write the various applications of HPLC.                                                                                                       |
| c. | Explain the biosynthesis of any four of the followings:<br>(i) Ephedrine (ii) Sennosides (iii) Quercetin (iv) Umbelliferone (v) Cucurbitacins |

**SECTION C**

**3. Attempt any five parts of the following:**

**7 x 5 = 35**

|    |                                                                                                                                                  |
|----|--------------------------------------------------------------------------------------------------------------------------------------------------|
| a. | Discuss the history of herbs as source of drugs & product discovery process.                                                                     |
| b. | Write a short note on supercritical fluid extraction.                                                                                            |
| c. | Discuss various methods of fractionations.                                                                                                       |
| d. | Explain the significance of HPTLC in characterization of herbal extract.                                                                         |
| e. | Explain the structure elucidation of any two of the followings:<br>(i) Kaempferol (ii) Glycyrrhizin (iii) Hecogenin.                             |
| f. | Write a short note on product discovery process & drug registration.                                                                             |
| g. | Discuss the structure & elucidation of citral & glycyrrhizin by spectroscopic techniques like UV, IR, MS, NMR ( <sup>1</sup> H, <sup>13</sup> C) |